Sales Forecasting and Demand Prediction

DEPI's Final Project Documentation

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Introduction

The Sales Forecasting and Demand Prediction project aims to develop a data-driven system capable of predicting future product sales based on historical patterns and external factors. Accurate sales forecasting plays an essential role in helping businesses make strategic decisions related to inventory management, resource allocation, marketing planning, and overall operational efficiency.

Many businesses struggle with accurately predicting how much of a product they will need in the future. Without reliable sales and demand forecasts, companies often face issues such as overstocking, stockouts, wasted resources, and missed opportunities to increase revenue. These challenges affect daily operations and may also lead to long-term financial losses. This project addresses these problems by developing a forecasting system that leverages historical sales data to better anticipate future demand.

The goal of the system is to support informed, data-driven decision-making related to:

- How much inventory to stock
- When to launch promotions
- How to allocate resources across different stores or regions
- How to optimize sales strategies for revenue growth

In this project, we apply data science and machine learning techniques to explore, analyze, and model the dataset. The workflow follows a structured milestone-based process, including data collection, exploratory analysis, feature engineering, model development, optimization, deployment, and monitoring. Each phase contributes to building a reliable forecasting model capable of capturing seasonality, trends, promotions, and other factors influencing demand.

By the end of the project, the model provides actionable insights that enhance business planning, reduce stockouts, prevent overstocking, improve customer satisfaction, and ultimately increase profitability. This documentation summarizes the full process, key decisions, and outcomes that led to the development of an effective and practical sales forecasting solution.

Project Overview and Business Impact

Throughout this project, our goal was to understand and predict sales and demand patterns in a way that genuinely helps businesses make smarter decisions. By carefully exploring the data, engineering meaningful features, and developing accurate forecasting models, we built a system that does more than just make predictions — it reveals the factors that drive customer behavior.

This work goes beyond numbers: it equips businesses with tools to prevent overstocking, avoid stockouts, better prepare for seasonal changes, and respond confidently to market shifts. While there is always room for improvement, such as integrating live data or automating updates, this project represents a solid step toward smarter, data-driven planning.

Milestone 1: Data Collection, Exploration & Preprocessing

1.1 Overview

- **Purpose of this phase:**

  The purpose of this phase is to gather a reliable sales dataset, analyze its structure, and prepare it for building forecasting models. This stage ensures that the data is clean, consistent, and enriched with meaningful features that support accurate prediction.

  Through data exploration and preprocessing, we aim to understand sales patterns, identify trends and seasonality, handle missing or inconsistent values, and transform the raw dataset into a high-quality foundation suitable for advanced analysis and model development.

1.2 Data Collection

- **File:** Superstore_Sales_Dataset.csv
- **Description:**
  The dataset includes transactional sales data, customer information, product categories, shipping details, and financial metrics such as profit and discount. This data is used to understand past sales behavior and forecast future demand across different products, regions, and customer segments.
- **Data fields table:**

Field Name	Data Type	Description
Row ID	Integer	Unique identifier for each row in the dataset.
Order ID	String	Unique code for each order.
Order Date	Date	Date when the order was placed.
Ship Date	Date	Date when the order was shipped.
Ship Mode	Category	Shipping method chosen (e.g., Standard Class, First Class).
Customer ID	String	Unique identifier for each customer.
Customer Name	String	Full name of the customer.

Segment	Category	Customer segment (Consumer, Corporate, Home Office).
City	String	City of the customer/order.
State	String	State of the customer/order.
Country	String	Country where the order was placed.
Postal Code	Integer	Postal code of the delivery location.
Market	Category	Market classification (e.g., APAC, EU, US).
Region	Category	Regional location of the order.
Product ID	String	Unique identifier for each product.
Category	Category	Main product category (Furniture, Office Supplies, Technology).
Sub-Category	Category	Sub-category of the product (e.g., Chairs, Binders, Phones).
Product Name	String	Full name of the product.
Sales	Float	Revenue generated from the order.
Quantity	Integer	Number of units sold.
Discount	Float	Discount applied to the order.
Profit	Float	Profit earned from the order.
Shipping Cost	Float	Cost required to ship the order.
Order Priority	Category	Priority level of the order (Critical, High, Medium, Low).

- Notes on data acquisition:
    1. The dataset was collected from a structured sales records file containing product, customer, and transaction-level details.
    2. **No APIs or web scraping** were used; the data is entirely **file-based**.
    3. The dataset is **suitable for time-series forecasting** due to the presence of **Order Date**, **Sales**, **Quantity**, and **Profit**.
    4. Additional preprocessing (date conversion, missing values, duplicate handling) will be required in later steps.

1.3 Data Exploration (EDA)

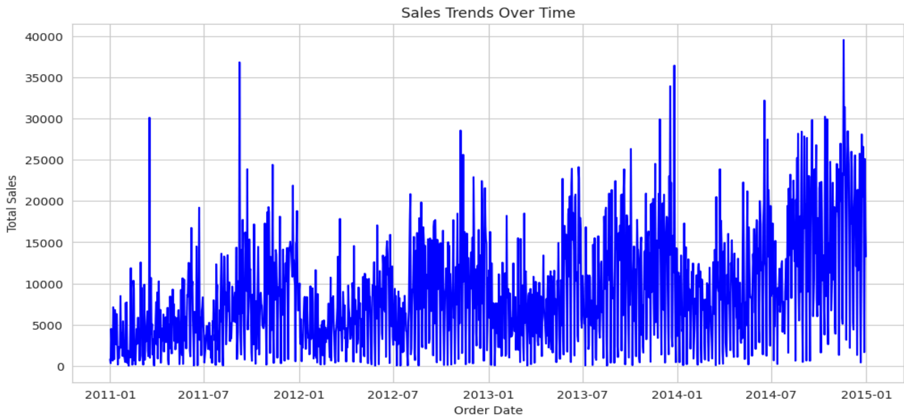
We began by exploring the dataset, which included historical records of product sales, store locations, discount rates, and several other business-related variables. The initial analysis helped us understand the distribution of data, identify missing values, detect outliers, and uncover patterns such as seasonal trends or relationships between variables like discounts and sales volume. Interactive visualizations were also created to offer dynamic views of key metrics and patterns across time and different stores or product categories.

- **Data Loading & Preprocessing:**
    - Imported dataset successfully after testing multiple encodings (`latin-1`, `iso-8859-1`, etc.).
    - Converted `Order Date` and `Ship Date` to datetime format for proper time-based analysis.
    - Removed unnecessary columns such as `Row ID`, `Order ID`, `Customer ID`, `Postal Code`, and `Product ID`.
    - Added new features:
        - Time-based: `Month`, `Quarter`, `Year`, `Day_of_Week`, `Is_Weekend`.
        - Holiday indicator: `Is_Holiday` based on US Federal Holidays.
        - Seasonal classification: `Season` (Winter, Spring, Summer, Fall).
        - Total sales: `Total_sales` calculated as `Sales × Quantity`.

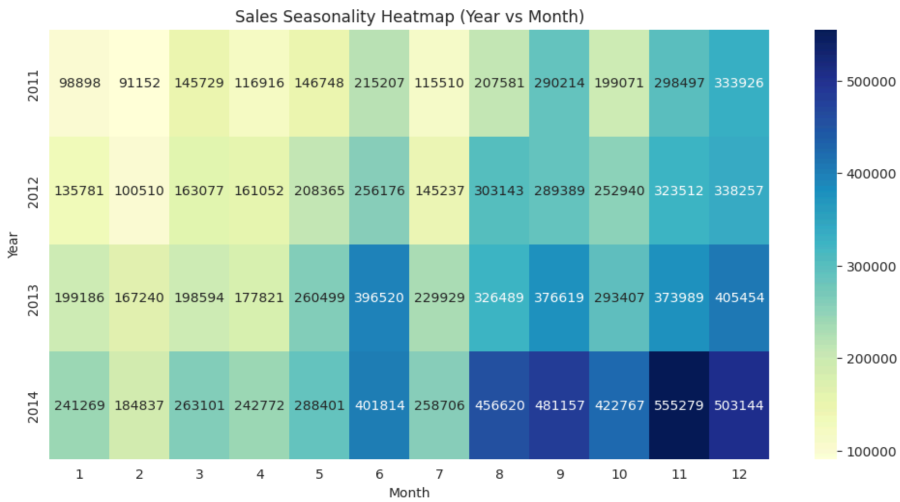
- **Basic Summary Statistics:**
    - Sales distribution by `Category` and `Sub-Category` highlighted top contributors (`Technology`, `Furniture`, `Office Supplies`) and best-selling sub-categories like `Phones`, `Copiers`, `Chairs`.Shipping mode distribution showed most orders used `Standard Class`.
    - Regional sales and profit analysis identified `Central` as the top region in both metrics.

Metric	Value
Number of records	51,290
Total Sales Revenue	$12,642,501.91
Total Profit	$1,467,457.29
Total Discounts Given	7,329.73
Average Sales per Order	$246.49

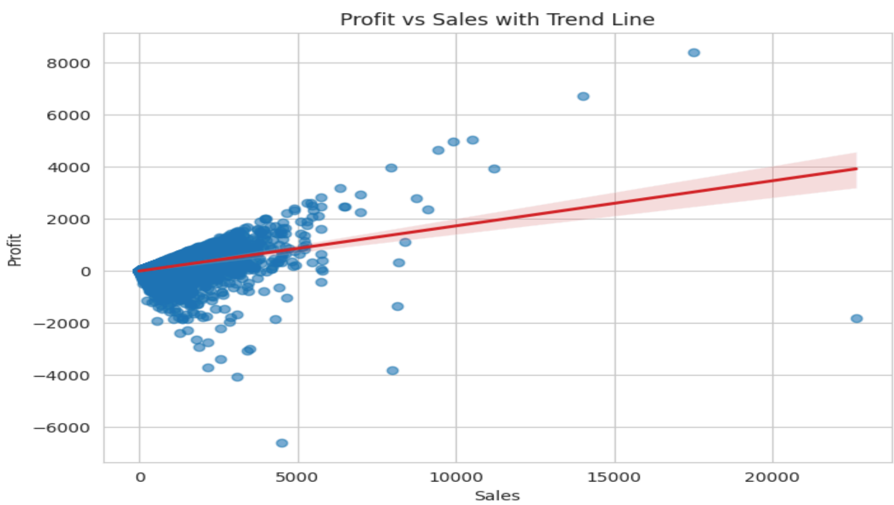
- **Sales Trends & Seasonality:**
  - Aggregated sales by `Order Date` to observe time series trends.



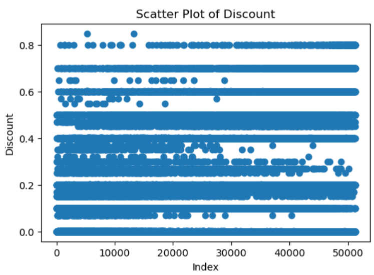
  - Monthly sales and profits analyzed by `Year` and `Month` to detect seasonality.

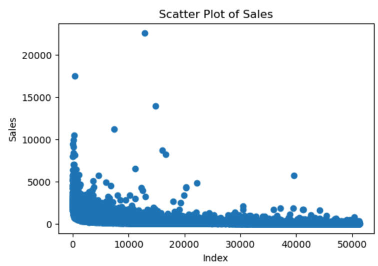


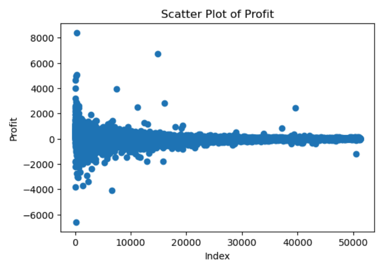
  - Relationship between `Sales` and `Profit` explored via scatter plot.

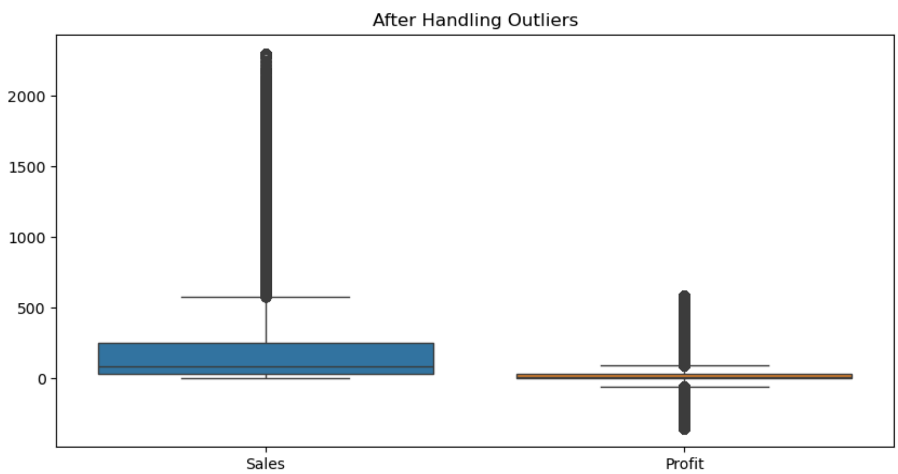


- **Customer & Product Insights:**
    - Top 10 customers by total sales identified.
    - Top 5 most profitable products vs least profitable products listed.
    - Top 5 most ordered vs least ordered products analyzed based on `Quantity`.
- **External Factors Impact:**
    - `Is_Holiday` and `Is_Weekend` features analyzed to assess impact on sales and profit:
        - Sales and profit are **slightly higher** on holidays and weekends.
    - **Average shipping cost** is analyzed by `Season` and `State` to identify peak logistics costs.
    - **Shipping performance** is measured using `Shipping Time (Days)` to find fastest and slowest deliveries.
- **Data Quality Checks:**
    - No missing values detected across all columns.
    - No duplicate rows found.
- **Outlier Detection**
    - Outlier detection was performed using **scatter plots** to visually identify extreme values in the numeric columns (`Sales`, `Discount`, `Profit`, `Shipping Cost`). The scatter plots helped highlight data points that deviate significantly from the general distribution.
        - Identified number of outliers in each column.





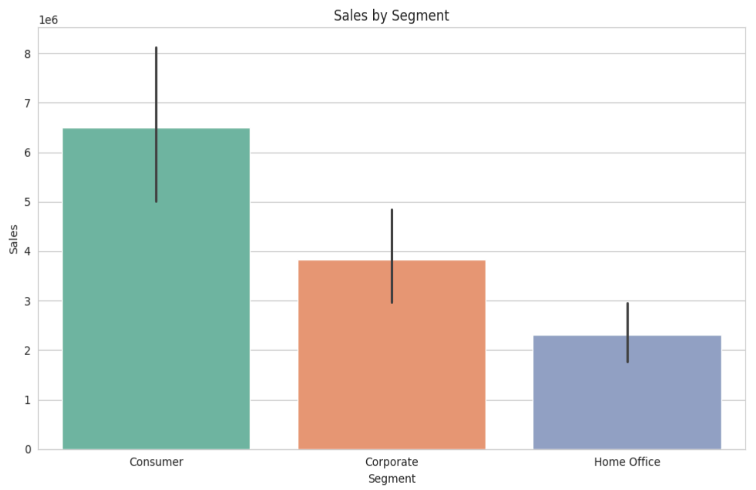


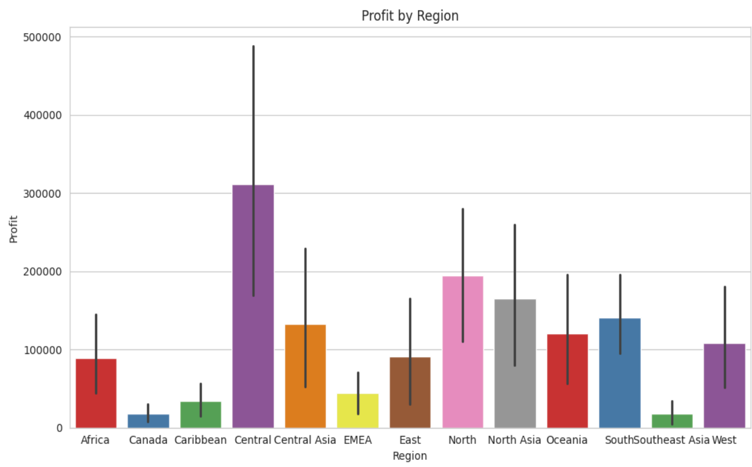


- **Key Columns Analyzed:**
  - **Numeric:** `Sales`, `Profit`, `Discount`, `Shipping Cost`, `Quantity`, `Total_sales`.
  - **Categorical:** `Category`, `Sub-Category`, `Product Name`, `Region`, `City`, `Ship Mode`, `Season`, `Is_Holiday`, `Is_Weekend`.

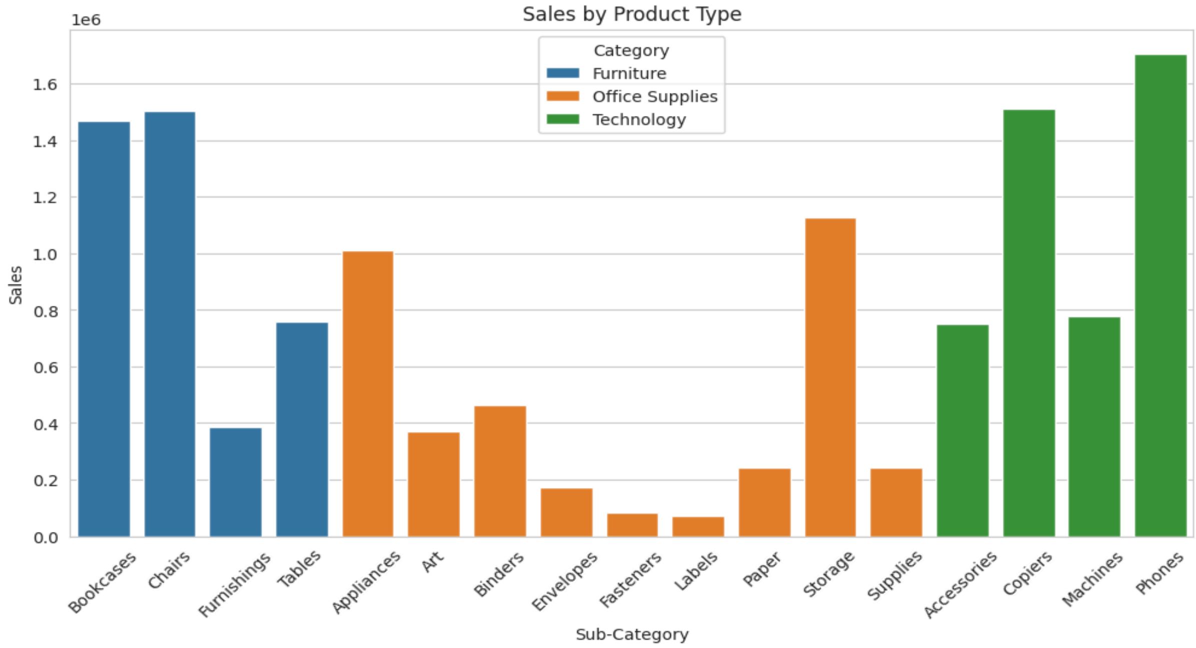
- **Basic Statistical Analysis**:
  - **Calculated summary statistics** (count, mean, standard deviation, min, max, quartiles) for all numeric columns.
  - **Additional metrics computed:** skewness, kurtosis, median, variance, mode, and range.
  - **Columns analyzed**: `Sales`, `Profit`, `Quantity`, `Discount`, `Shipping Cost`.
  - **Result:** Achieved comprehensive understanding of central tendency, dispersion, and distribution of numeric features.

- **Aggregated Analysis by Region, Market, Segment, and Order Priority**:
  - Total `Sales`, `Profit`, and `Quantity` computed for each combination.
  - Identify top-revenue regions, markets, or segments.

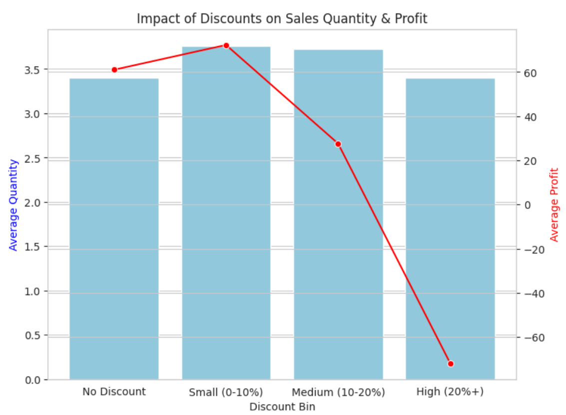


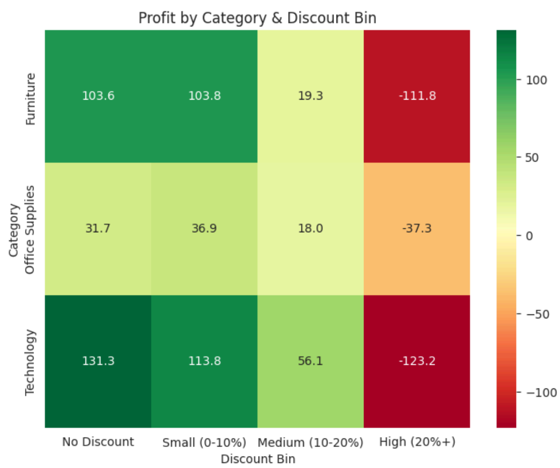


- **Sales & Profit by Product Type**:
  - Grouped data by `Category` and `Sub-Category` to compute total `Sales`, `Quantity`, and `Profit`.
  - Profit Margin calculated as `Profit / Sales`.
  - **Insights:** Identified sales distribution and top profitable sub-categories.

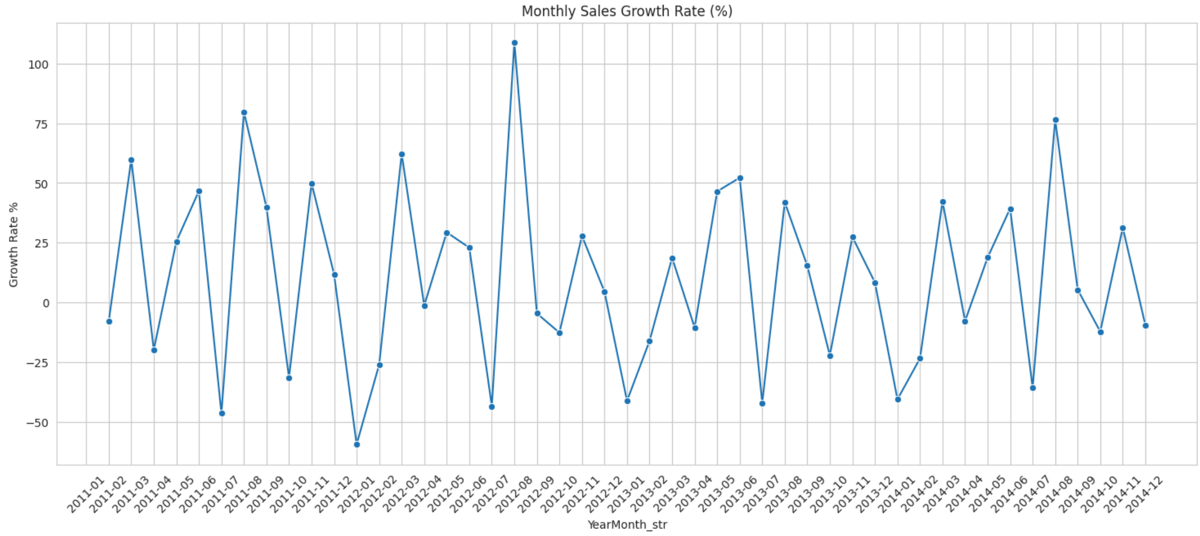


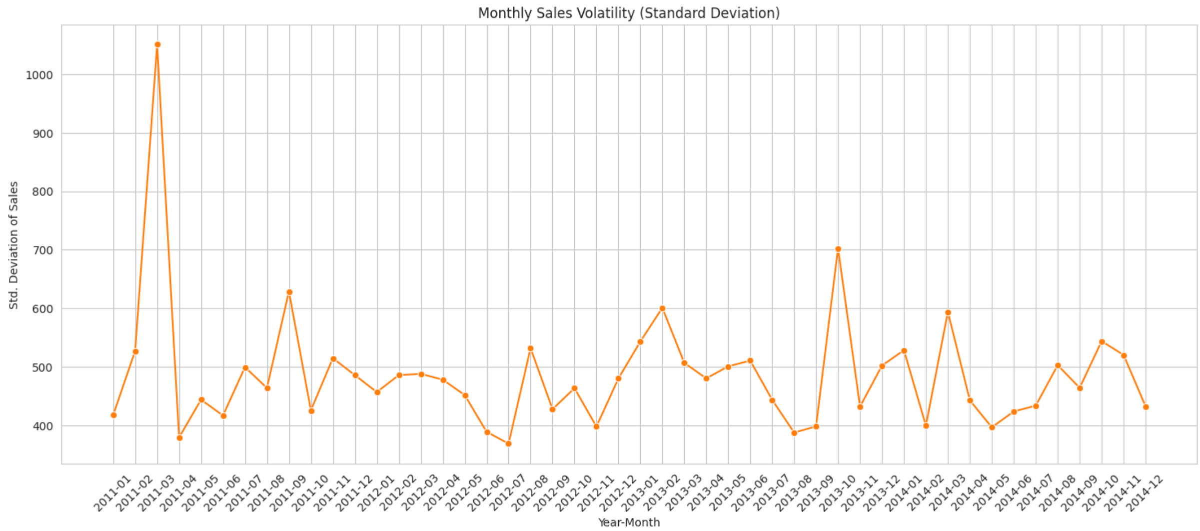
- **Impact of Discounts on Sales and Profit**:
  - Created `Discount Bin` feature (No Discount, Small, Medium, High).
  - Analyzed average `Quantity` sold and `Profit` for each discount level.
  - Observed trends: Medium discounts often balance higher sales quantity with profit.
  - Heatmap created to visualize average profit per `Category` across discount bins.



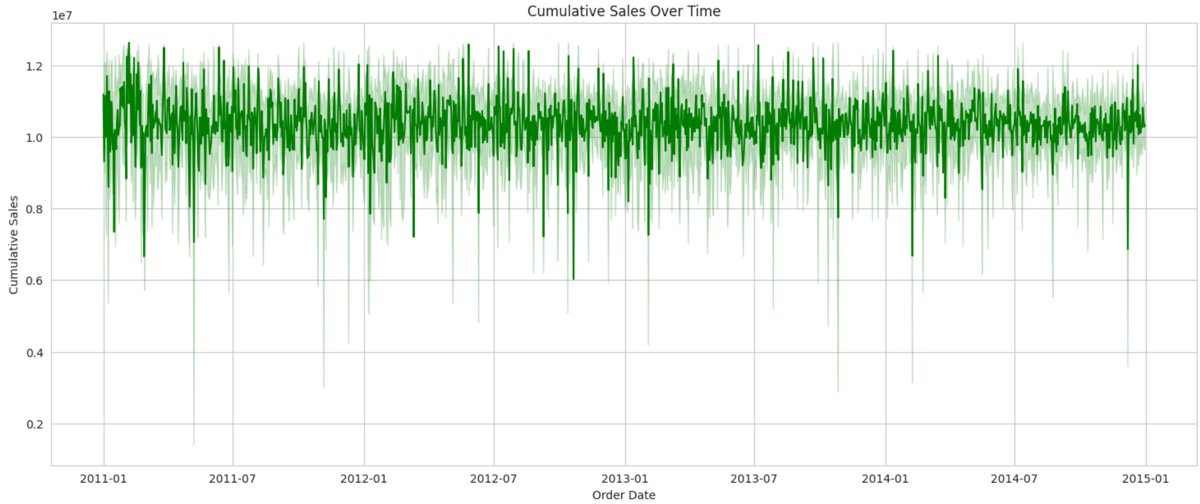


- **Monthly Sales Growth & Volatility**:
    - Aggregated sales by `YearMonth` to calculate monthly sales growth rates (% change).
    - Volatility computed using standard deviation of sales per month.
    - Insights: Identify months with high growth or high fluctuation.

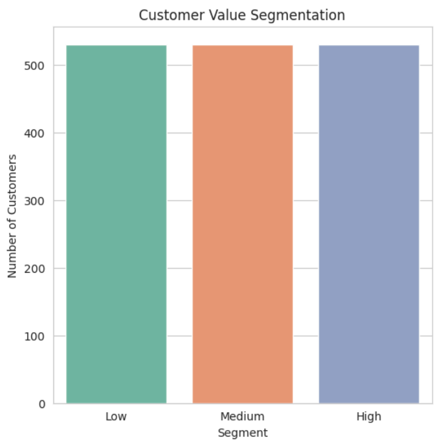


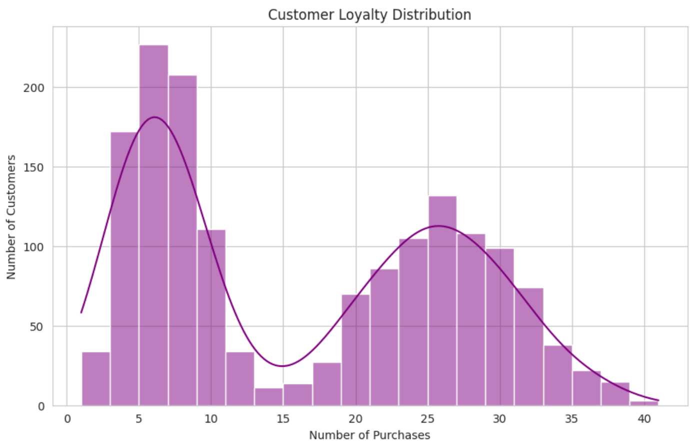


- **Cumulative Sales Over Time**:
  - Calculated cumulative sum of `Sales` over time to visualize overall sales accumulation.
  - Steeper slopes indicate periods of strong sales, flatter slopes show low activity.



- **Customer Analysis**:
  - **Value Segmentation**: Customers divided into Low, Medium, High based on total sales.
  - **Loyalty Analysis**: Counted number of purchases per customer to evaluate repeat buying behavior.
  - **Basket Size Analysis**: Average number of products per order computed.
  - Insights: Understand which customers contribute most, frequency of purchases, and typical order size.





- **Time-based Sales Patterns**:
    - Extracted `Hour` and `DayOfWeek` from `Order Date` to analyze hourly and weekday sales trends.
    - Pivoted data to create a heatmap showing sales distribution across days and hours.
    - Insights: Identify peak hours and days for sales.

- **Key Columns Summarized in EDA**:
    - `Sales`, `Profit`, `Quantity`, `Discount`, `Shipping Cost`, `Category`, `Sub-Category`, `Segment`, `Region`, `Market`, `Order Priority`, `Customer ID`, `Order Date`, `Hour`, `DayOfWeek`, `Cumulative Sales`, `Discount Bin`, `Profit Margin`, `YearMonth`.

- **General Insights**:
    - Sales trends, seasonality, and product performance are clearly visualized.
    - Customers segmented effectively for targeted marketing strategies.
    - Discount strategies and shipping patterns evaluated for business decisions.
    - Identified temporal patterns to optimize sales and inventory management.

1.4 Preprocessing

- **Dataset Loading and Initial Inspection**
  The dataset was loaded and examined to understand its structure, size, and data types. Summary statistics and the first few rows were reviewed.

  - **Key Findings:**
    - **Total records:** 51,290
    - **Total features:** 24
    - **Data types:** 17 categorical features and 7 numerical features
    - Significant missing values identified in **Postal Code** (≈ 41K missing)

- **Exploration of Categorical Features**
  Unique values were extracted for all categorical columns to assess variability and distribution.

  - **Observations:**
    - **Country:** 147 unique values
    - **City:** 3,636 unique values
    - **State:** 1,094 unique values
    - **Order Priority:** Four levels → *Critical, High, Medium, Low*
    - **Category:** 3 major categories
    - **Sub-Category:** 17 sub-categories

  This helped identify high-cardinality features and structure the encoding strategy.

- **Removal of Irrelevant or Non-Informative Columns**
  The following columns were dropped due to irrelevance, high cardinality, or their nature as identifiers:
  (`Postal Code` - `Row ID` - `Customer Name` - `Customer ID` - `Order ID` - `City` - `State`)

  - **Reasoning:**
    These features do not contribute to predictive performance and introduce noise.
    Post-removal, the dataset contained **17** features.

- **Outlier Detection (IQR Method)**
  Outliers were identified using the Interquartile Range (IQR) method for key numerical features:
  1. **Sales:**
     Outliers detected: **5,655**
     Acceptable Range: **−299.68** to **581.50**

  2. **Profit**
     Outliers detected: **9,755**
     Acceptable Range: **−55.22** to **92.03**

  3. **Discount**
     Outliers detected: **4,172**
     Acceptable Range: **−0.30** to **0.50**

- **Outlier Treatment**
  Outliers were handled using appropriate domain-aligned strategies:
  - **Sales & Profit:**
    Winsorization (**1st–99th** percentile capping).
    - Sales range adjusted to: **3.69** to **2301.00**
    - Profit range adjusted to: **−351.51** to **587.36**
  - **Discount:**
    Values outside the logical range (0–1) were removed.

- **Date Conversion and Cleaning**
  Order Date and Ship Date were converted into proper datetime format.

Date Conversion and Cleaning

```
df['Order Date'] = pd.to_datetime(df['Order Date'], format='%d-%m-%Y')
df['Ship Date']  = pd.to_datetime(df['Ship Date'], format='%d-%m-%Y')
```

- **Feature Engineering (Time-Based Attributes)**
  Several date-derived features were created to support seasonality and trend analysis.
  1. **Order-Related Features:**
     (`Order_Year` - `Order_Month` - `Order_Day` - `Order_Weekday` - `Order_Is_Weekend` - `Order_Week` - `Order_Quarter`)

  2. **Shipping-Related Features:**
     The same components were extracted for the Ship Date.

  3. **Shipping Duration:**
     A new feature (*`Shipping_Days`*) was computed:
     `Shipping_Days = Ship Date − Order Date`

- **Encoding Categorical Features**
    - **Label Encoding (Order Priority)**
      Customized ordinal mapping was applied:

Priority	Encoded Value
Critical	4
High	3
Medium	2
Low	1

    - **One-Hot Encoding**
      Performed for:
      (`Country` - `Region` - `Market` - `Category` - `Sub-Category` - `Ship Mode`)

      Using:

      One-Hot Encoding
      ```
      df = pd.get_dummies(df, columns=one_hot_cols, drop_first=True)
      ```

- **Scaling Numerical Variables**
- **Min-Max** normalization was applied to ensure that numerical features operate on a comparable scale, especially for machine learning models.
    - **Scaled Features:**
      (`Sales` - `Quantity` - `Discount` - `Profit` - `Shipping Cost` - `Shipping_Days`)
    - **Final Output**
      The dataset after **EDA** and preprocessing is:
        - Clean
        - Free of invalid and extreme values
        - Fully encoded
        - Enriched with engineered features
        - Properly scaled

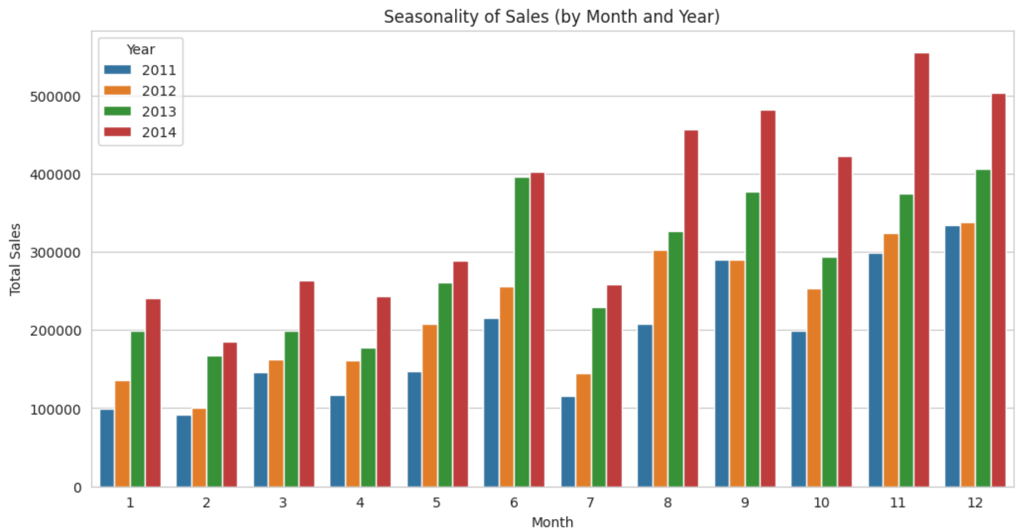
- **Key Insights from EDA for Forecasting (Summary)**

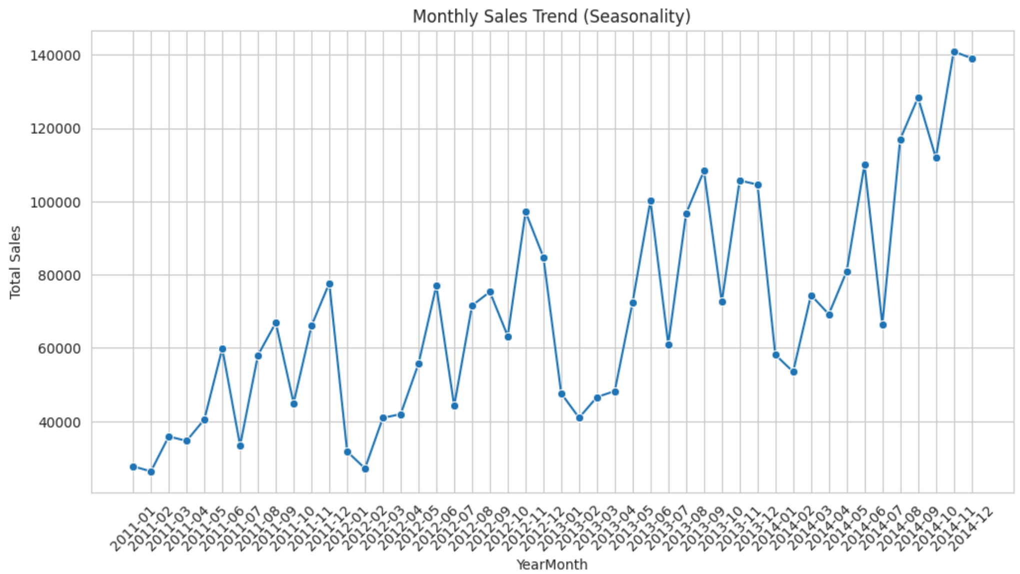
  The exploratory analysis revealed clear seasonality and noticeable month-to-month volatility in sales, indicating the need for time-series models that capture cyclic patterns. Significant differences were observed across regions, segments, and product categories, confirming that these features strongly influence demand and profitability. Discount levels also showed a consistent impact on both quantity sold and profit, highlighting the importance of price-sensitivity in forecasting.
  Customer behavior patterns—including value segmentation, purchase frequency, and basket size distribution—displayed meaningful variation that can enhance prediction accuracy. Overall, the EDA provided essential insights that guide feature selection and support the development of reliable forecasting models.

Milestone 2: Advanced Data Analysis and Feature Engineering

2.1 Advanced Analysis

- **Date & Time Feature Extraction**
  To enhance the temporal understanding of sales behavior, multiple time-based features were engineered from the Order Date:
  1. Converted `Order Date` and `Ship Date` into standard DateTime format.
  2. Extracted key calendar components including `Month`, `Week Number`, `Day of Month`, and `Day of Week` (0 = Monday, 6 = Sunday).
  3. Assigned a **Season** label (`Winter`, `Spring`, `Summer`, `Fall`) based on the month to detect seasonal patterns.
  4. Added a **Holiday flag** indicating whether the order occurred on major holidays (New Year or Christmas).



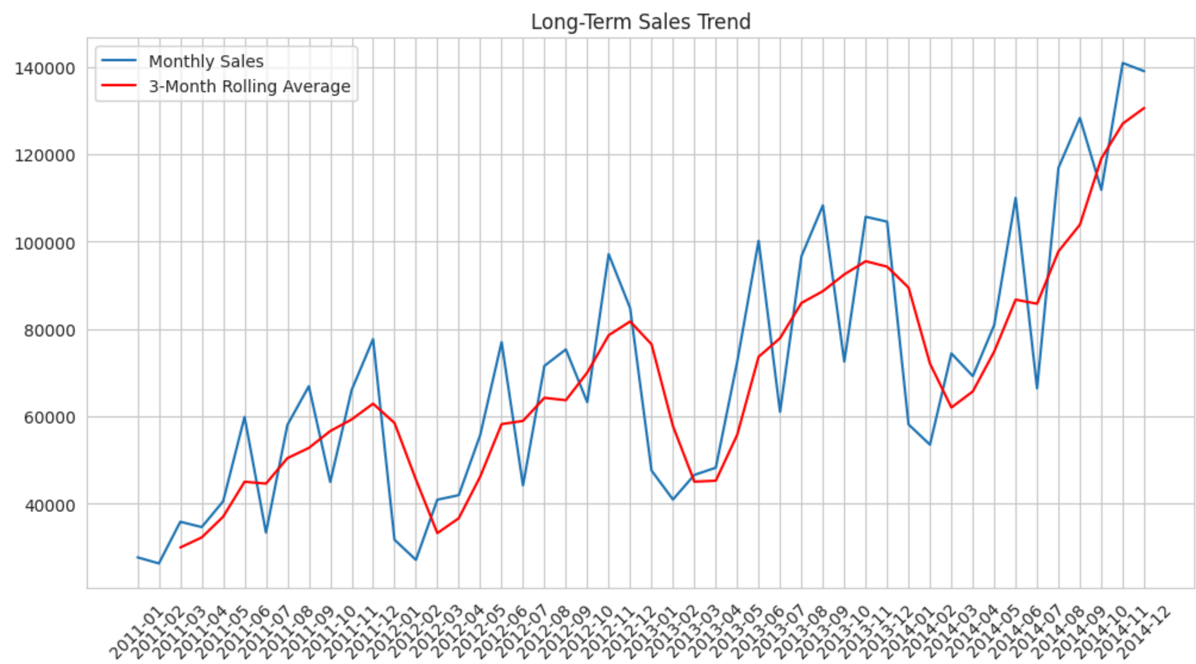


2.2 Advanced Feature Engineering

- **Lag Features for Temporal Dependency Modeling**
  To capture the relationship between past and future values in forecasting models, several lag features were constructed:
  - **Sales_Lag_1D**: sales value from the previous day.
  - **Sales_Lag_7D**: sales value from the previous week.
  - **Sales_Lag_30D**: sales value from the previous month.
  - **Profit_Lag_7**: profit recorded 7 days prior (used backfill to avoid missing values).
  - **Shipping_Lag_14** and **Shipping_Lag_30**: shipping cost values from two and four weeks earlier.

  Notable Output Observation:
  - Shipping cost showed **no variation over 30 days**, indicating a static pricing policy.
  - Weekly profit lags revealed consistent short-term fluctuations.

- **Rolling Statistical Features**
  To detect medium-term smoothing patterns in sales and profit:
  - **7-Day Rolling Average for Sales**:
    Provided a smoothed short-term trend, reducing day-to-day noise.
  - **30-Day Rolling Average for Profit**:
    Highlighted monthly profitability stability and trend changes.



- **Additional Enhanced Calendar Features**
  For more granular demand pattern analysis:
  - `Month Name`, `Day Name`, and `Quarter` features were added.
  - Reconstructed a refined seasonal mapping for consistency across all processed data.

  Summary of Key Outputs
  1. Seasonal and holiday-based features revealed **minimal sales difference across seasons** with expected peaks around holidays.
  2. Lag features successfully captured **short-term and weekly dependencies**, essential for improving forecasting accuracy.
  3. Rolling averages and EWMA offered **smoother, more interpretable trends**, supporting long-term demand analysis.
  4. Shipping cost lags showed **nearly constant values**, indicating that shipping expense is not a major driver of sales variability.

Milestone 3: Model Development & Optimization

In this stage, multiple regression models were developed and evaluated to predict **Total Sales** based on the engineered features from the preprocessing phase. The dataset was split into training and testing subsets (80/20) to ensure unbiased performance assessment.

3.1 Model Training Overview

Four machine learning models were trained and compared:

- **Linear Regression**
- **Random Forest Regressor**
- **Gradient Boosting Regressor**
- **XGBoost Regressor**

Each model was evaluated using **RMSE**, **MAE**, and **R^2** on both **training** and **testing** sets to assess **accuracy** and **generalization**.

3.2 Model Performance Summary

Training Set Evaluation:

Model	RMSE	MAE	R^2
Linear Regression	289.05	191.22	0.6957
Random Forest Regressor	72.62	39.78	0.8288
Gradient Boosting Regressor	202.62	122.03	0.8756
XGBoost Regressor	149.86	85.92	0.9454

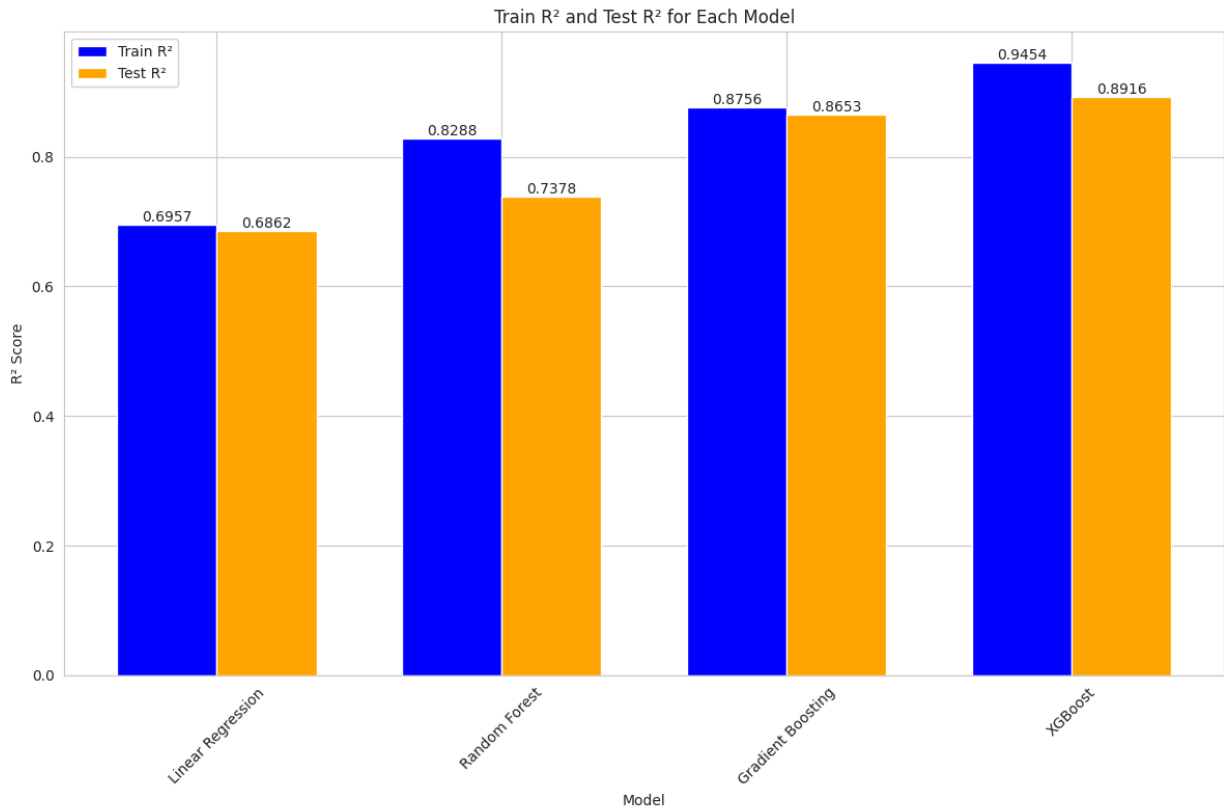
Test Set Evaluation:

Model	RMSE	MAE	R^2
Linear Regression	288.09	288.09	0.6862
Random Forest Regressor	192.59	106.28	0.7378
Gradient Boosting Regressor	206.00	123.51	0.8653
XGBoost Regressor	194.10	108.58	0.8916

3.3 Model Comparison & Selection

Performance summary:

- **R^2 Comparison Across Models**
  A grouped bar chart was generated to compare training and testing R^2 scores for all models.



Best-performing model:
Hyperparameter Tuning (XGBoost)
A comprehensive Grid Search was performed on XGBoost using 5-fold cross-validation across **2,916 parameter combinations**.

Best Parameters Identified

```
GridSearchCV

Best parameters: {
            'colsample_bytree': 0.8,
            'gamma': 0.1,
            'learning_rate': 0.05,
            'max_depth': 7,
            'n_estimators': 300,
            'reg_alpha': 1,
            'reg_lambda': 2,
            'subsample': 1.0
}
```

```
GridSearchCV

Tuned XGBoost RMSE: 187.07
Tuned XGBoost MAE: 103.56
Tuned XGBoost R²: 0.8173
```

Key Insights

- Ensemble models (Random Forest, Gradient Boosting, XGBoost) significantly outperformed Linear Regression.
- **Random Forest** achieved the highest training accuracy but showed a notable gap between training and testing errors, indicating **overfitting**.
- **XGBoost and Gradient Boosting** demonstrated better generalization with strong test R^2 and controlled overfitting.
- The tuned XGBoost model provided the **best balance between accuracy and robustness**, making it the strongest candidate for forecasting tasks.

Milestone 4: MLOps, Deployment & Monitoring

4.1 Dashboard Deployment

A **Streamlit-based interactive dashboard** was implemented to facilitate **real-time sales prediction** and data exploration:

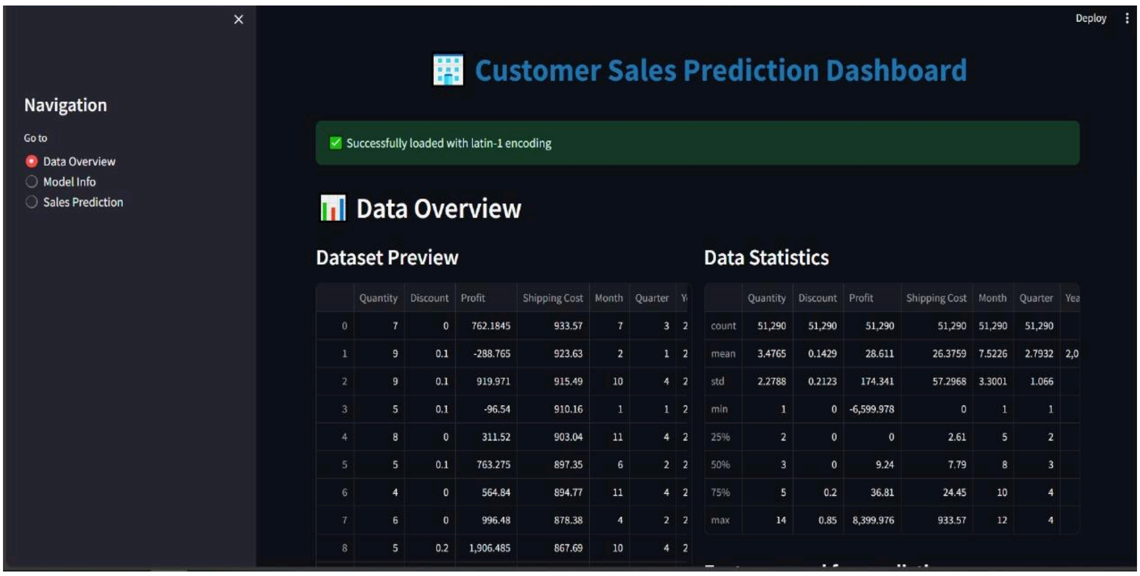
- **Page Configuration**: Set with wide layout, custom page title, and icon.
- **Custom Styling**: Applied CSS to enhance readability and visually differentiate prediction categories (High, Medium, Low Sales).
- **Navigation**: Sidebar for switching between **Data Overview**, **Model Info**, and **Sales Prediction** pages.

4.2 Data Loading & Preprocessing

The dashboard integrates a robust data ingestion pipeline:

1. **Encoding Detection**: Automatically detects CSV file encoding to ensure accurate data loading.
2. **Date Conversion & Feature Engineering**: Order and Ship dates converted to DateTime; additional temporal features generated: Month, Quarter, Year, Day of Week, Weekend flag, and holiday indicator.
3. **Promotion & Season Features**: Created flags for discounts and mapped months to seasons.
4. **Label Encoding**: Categorical variables transformed to numeric format for model compatibility.
5. **Missing Value Handling**: Numerical columns imputed with mean values; categorical columns imputed with mode or 'Unknown'.

Key Output:
Cleaned dataset ready for prediction with engineered features.



4.3 Model Integration & Monitoring

The dashboard supports **XGBoost model deployment**:

- Pre-trained model loaded via pickle.
- Ensures correct **feature order** for predictions to prevent misalignment.
- Displays **model information**: type, feature importance, and top 5 contributing features.

Key Output:
Feature importance chart highlights the most influential features affecting total sales prediction.

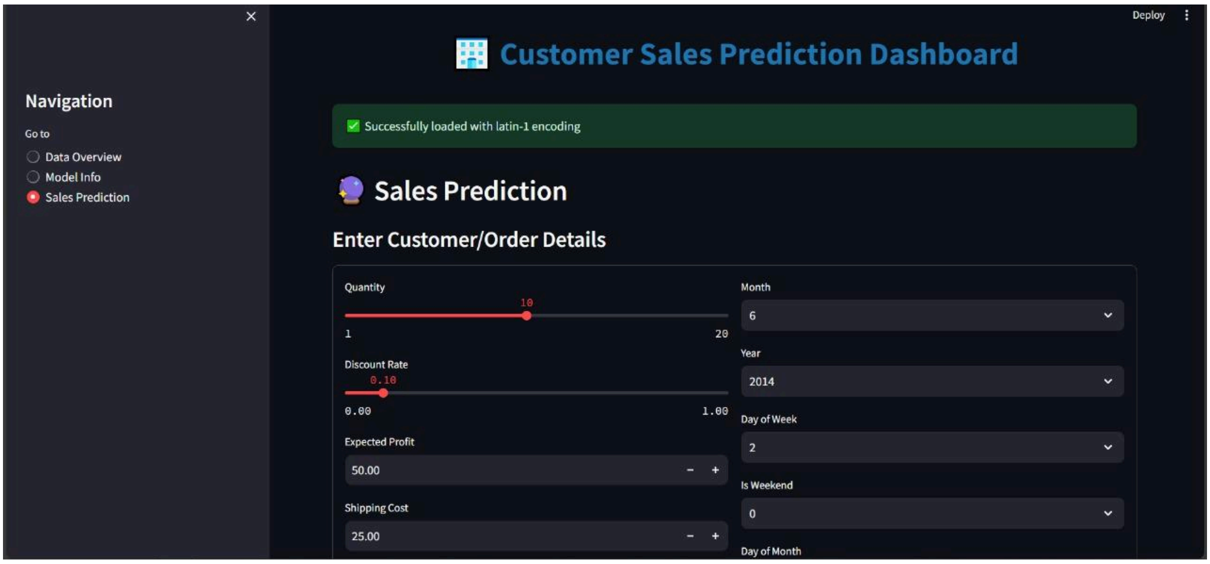
4.4 Interactive Sales Prediction

Users can input **customer and order details** to predict total sales:

- Input widgets include: Quantity, Discount, Profit, Shipping Cost, Month, Year, Day of Week, Weekend flag, Ship Mode, Market, Category, Sub-Category, and Order Priority.
- Predictions categorized into **High**, **Medium**, and **Low Sales** based on dataset quantiles.
- It displays results dynamically with visual emphasis using custom CSS.

Key Output:

- Predicted sales for user-provided input with visual classification.
- Alerts provided when default values are used for missing features.



Summary of Key Outcomes

- **End-to-End MLOps Workflow**:
  Includes data ingestion, preprocessing, feature engineering, model integration, prediction, and monitoring.
- **User-Friendly Dashboard**:
  Provides actionable insights without technical expertise.
- **Real-Time Predictions**:
  Facilitates business decision-making by forecasting sales for specific orders.
- **Monitoring & Maintainability**:
  Cached data and model loading ensure performance and reproducibility; warning/error messages guide users in case of issues.